Chapter 4

ProD: A Tool for Predictive Design of Tailored Promoters in *Escherichia coli*

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Abstract

A major goal in synthetic biology is the engineering of synthetic gene circuits with a predictable, controlled and designed outcome. This creates a need for building blocks that can modulate gene expression without interference with the native cell system. A tool allowing forward engineering of promoters with predictable transcription initiation frequency is still lacking. Promoter libraries specific for σ^{70} to ensure the orthogonality of gene expression were built in *Escherichia coli* and labeled using fluorescence-activated cell sorting to obtain high-throughput DNA sequencing data to train a convolutional neural network. We were able to confirm in vivo that the model is able to predict the promoter transcription initiation frequency (TIF) of new promoter sequences. Here, we provide an online tool for promoter design (ProD) in *E. coli*, which can be used to tailor output sequences of desired promoter TIF or predict the TIF of a custom sequence.

Key words Synthetic biology, Machine learning, Promoter design, Metabolic engineering, Biotechnology, Toolbox

1 Introduction

As metabolic engineering advances in both fundamental knowledge on the functioning of microorganisms as well as in providing tools to achieve precise engineering goals, the required level of precision increases simultaneously [1–3]. To improve current production titers by engineering a desired expression level, it is often not sufficient to find and turn the right screw. Combinatorial pathway optimization has been one strategy to reduce the time invested in searching for the optimal pathway which maximizes production [4, 5]. In general, control can be executed on three levels: transcriptional, translational and post-translational. While libraries exist

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Eveline Peeters and Indra Bervoets (eds.), *Prokaryotic Gene Regulation: Methods and Protocols*,
Methods in Molecular Biology, vol. 2516, https://doi.org/10.1007/978-1-0716-2413-5_4,

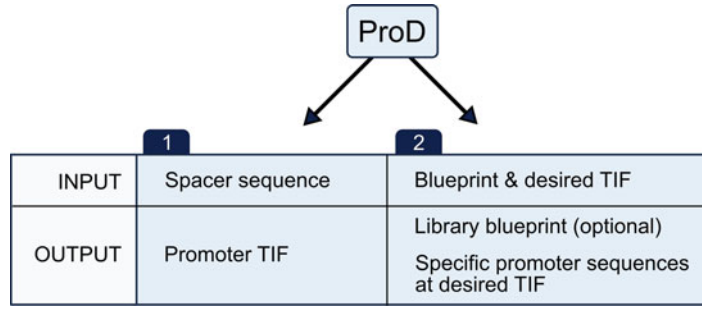

Fig. 1 The two-way usage of ProD, a promoter transcription initiation frequency (TIF) prediction tool supporting de novo design of promoter sequences. ProD can perform 2 types of predictions: In (1), the tool takes a custom spacer sequence and predicts the TIF. In (2), an input blueprint and desired TIF are used to output the blueprint for a library at the desired TIF, as well as specific spacer sequences for each requested TIF, or a single spacer sequence for each requested TIF

with promoters and ribosome binding site (RBS) sequences at predefined strengths [6], such libraries lack the flexibility that might be required in new setups. The success of the RBS (Library) Calculator from the Salis lab [7, 8] has shown the high demand of the synthetic biology community for such predictive tools that output a de novo library. In this work, we present ProD, a promoter transcription initiation frequency (TIF) prediction tool supporting de novo design of promoter sequences. This tool was built using promoters with randomized spacer regions in *Escherichia coli* that remained orthogonal by being specific to sigma factor σ^{70}. Orthogonality was ensured by leaving the -35 and -10 boxes adjacent to the fully randomized spacer region intact [9]. Data was collected by cloning the promoters upstream a fluorescent reporter gene, fluorescent-activated cell sorting (FACS) and subsequent DNA sequencing, allowing for the collection of considerably large datasets (250,000–400,000 unique sequences per setting) that mapped each sequence to fluorescence output. A custom-built convolutional neural network for ordinal regression was then trained on this dataset which is able to evaluate a custom spacer sequence, returning an estimated promoter TIF (0–10) based on the DNA sequence. Alternatively, ProD is able to output a promoter spacer sequence library at desired TIF or a library with specific promoter sequences in the range of the desired TIF (0–10, or very low to very high expression) [10]. An overview of the two workflows is depicted in Fig. 1.

2 Materials

ProD is available at https://github.com/MEMO-group/ProD and written in Python 3.7. The tool can either be run through Binder, which does not require a local installation, or installed on a local machine (*see* **Note 1**). To ensure compatibility in the latter case, the required package versions are listed in the repository.

2.1 Relevant Repository Files

1. ProD.py: the actual python script for the predictive promoter tool (*see* **Note 2**).
2. ProD_Notebook.ipynb: a Jupyter Notebook used for running code in an interactive environment. Contains detailed instructions on the functionality of the tool.
3. README.md: markdown readme file containing instructions on how to use the tool.
4. environment.yml: contains the package versions with which the tool was built. This custom environment ensures smooth running of the program. For ProD, Conda is used for environment management and therefore provided as a YAML file.
5. ex_seqs.fa: example input file used in the README.MD file. The fasta file is read as input and results in an output prediction csv file containing the predicted TIF for the input sequences (in this repository, it is called my_predictions.csv).

2.2 Prerequisites If ProD Is Run from the Terminal

1. Clone the ProD GitHub repository in your working directory (can also be downloaded as ZIP file from the website).
2. For Linux users, an environment can be created using the environment.yml file provided in the repository. For Mac and Windows users, verify that required Python libraries are available in your active environment:
   (a) Pytorch 1.4.0
   (b) Numpy 1.19.1
   (c) Pandas 1.1.0
   (d) Scipy 1.5.2

2.3 Prerequisites If ProD Is Utilized Using Jupyter

1. Install Jupyter Notebook or Jupyter Lab (https://jupyter.org).
2. Open ProD_Notebook.ipynb with Jupyter
3. A brief explanation on how to use the file is described in the file. In short, a code cell can be run by clicking on it and pressing Ctrl + Enter.

3 Methods

3.1 The ProD Tool

The ProD tool's motor is a convolutional neural network trained on 284,421 unique promoter sequences mapped to a fluorescent output value, which is representative for promoter TIF (*see* **Note 3**). This vast library of fluorescent output was achieved by introducing a spacer library. The spacer is framed by the −10 and −35 boxes to ensure an orthogonal promoter-σ^{70} pair. Davis et al. [11] have shown that by using an insulator sequence up- and downstream of the promoter element, the latter's TIF can be reliably predicted with minimal influence of the genetic context, both on plasmid and genome level. Hence, the promoter library was created using an insulator sequence as described by Davis et al. [11], containing a fixed transcription start site (UP sequence) upstream of the promoter, which facilitates polymerase binding, and a fixed untranslated region (UTR) downstream, facilitating promoter escape. The design of the insulator sequence is shown below:

***GGTCTATGAGTGGTTGCTGGATAAC*TTTACG**
(NNNNNNNNNNNNNNNNN)TATAATATATT
CA*GGGAGAGCACAACGGTTTCCCTCTACAAATAA*
TTTTGTTTAACTTT

The promoter region (−60 to +1) is shown in bold; the conserved −35 and −10 box are bold underlined and the UP region bold italic. The UTR is shown in italics. N_{17} represents the spacer region, whose sequence is predicted by the ProD tool (*see* **Note 4**). The TIF of an input spacer sequence, given the usage of the same conserved -35 and -10 boxes, can be predicted. The tool will be most accurate when used in the same insulator context, as Davis et al. [11] have shown the influence of surrounding regions on minimal promoters. When a promoter library is constructed (*see* **Note 5**), the tool outputs the entire insulator sequence as described above, as well as the spacer sequence, which can subsequently be assembled and cloned upstream an RBS and ORF of choice (*see* **Note 6**). The tool accepts blueprints to accommodate constraints in spacer design and takes 17 base pairs as input (*see* **Note 7**). The TIF is represented in classes from 0 to 10, with 0–2 = zero to low expression, 3–5 = low to medium expression, 6–8 = medium to high expression and 9–10 = high to very high expression. Figure 2 represents the workflow in the library setting.

3.2 Construction of Promoters with Desired TIF

The outcome of the tool is a mixed-base spacer sequence called library blueprint, as well as discrete spacer sequences for each requested TIF (0–10) and the resulting sequence of the entire insulator. Alternatively, a single spacer sequence at a specific TIF can be calculated. The generated csv file contains, in addition, the probability values of the spacer for each TIF class. A specific spacer

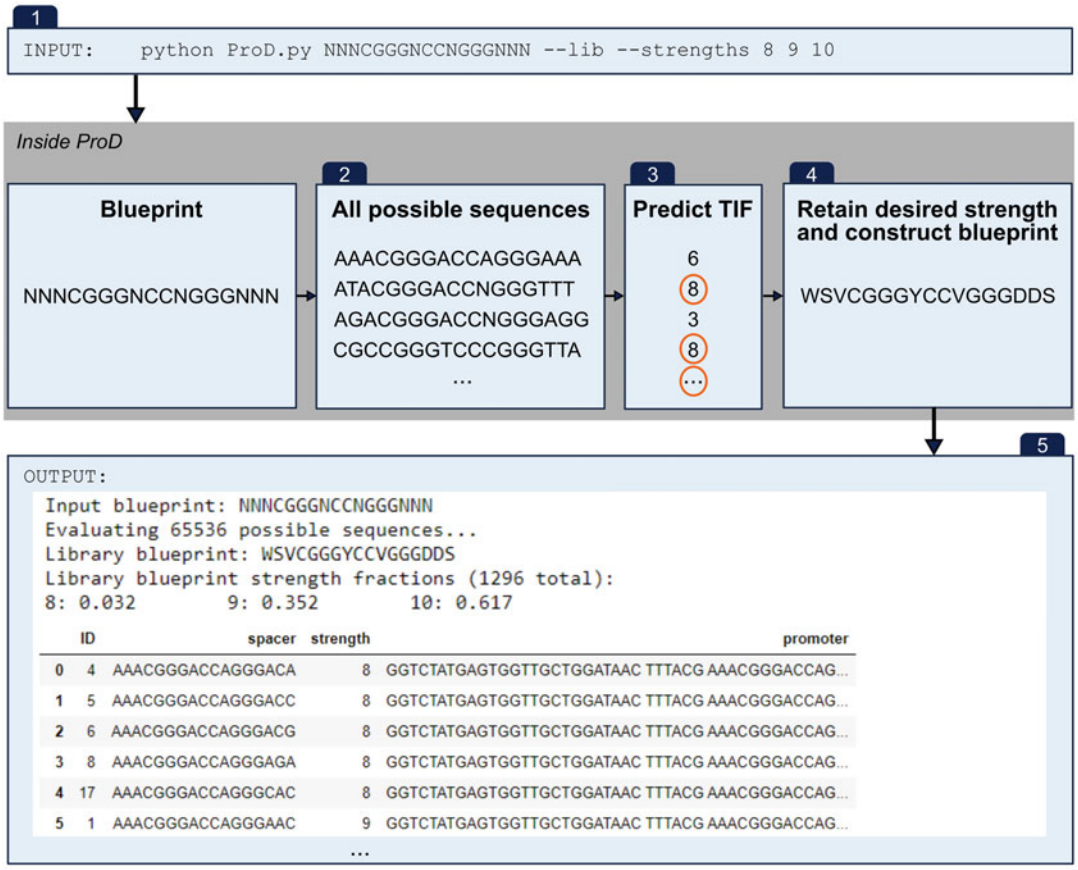

Fig. 2 The different steps in the process through which the ProD model obtains sequences with desired transcription initiation frequency (TIF). When constructing a library, the input (1) consists of a blueprint and desired TIF. The tool evaluates all possible sequences (2) and predicts their respective TIF (3). Only sequences of the desired TIF are retained from the previous step and a blueprint is constructed (4). The output (5) consists of the library blueprint, the fraction of sequences classified to each category of strength and a table with the spacer, TIF and insulator sequence. The CSV file my_predictions.csv further contains the probability for each TIF class for each spacer sequence

sequence or the library blueprint may be selected for subsequent cloning. The process through which the model obtains the final outputs is depicted in Fig. 2.

3.2.1 Using the Terminal

1. Open the terminal, activate the environment containing the packages described in Material.
2. Navigate to the directory containing the cloned GitHub repository.
3. Call the ProD tool inserting the desired library backbone and TIF(s), for example: python ProD.py NNNCGGG NCCNGGGNNN --lib --strengths 8 9 10. Alternative to a blueprint sequence, a fasta file containing a single blueprint

can be used as input: python ProD.py my_blueprint.fa --lib --strengths 8 9 10. If a single, specific spacer sequence is desired, add --lib_size 1.

4. The output is displayed in the terminal and additionally, a file my_predictions.csv is saved in the working directory. If another output path is desired, it can be specified in (3) by adding --output_path PATH

3.2.2 Using Jupyter

1. Open Jupyter in the environment containing the packages specified in Material and navigate to the directory containing the GitHub repository or load the ProD GitHub environment using the Binder cloud service.
2. Run the cell by selecting it and pressing Ctrl + Enter. The tool will be imported.
3. Enter your custom blueprint sequence in the next code cell as a string of 17 bases. Alternative to a blueprint sequence, a fasta file containing a single blueprint can be used as input: input_data = ["my_blueprint.fa"].
4. Define the TIFs in my_strengths as comma-separated integers. If a single, specific spacer sequence is desired, add lib_size=1 as argument of run_tool. Run the cell.
5. The output will be generated below the cell. This might take a while.
6. The generated csv file can be found in the working directory. If another output path is desired, it can be specified in **step 3** of this section by adding output_path=PATH as argument for run_tool.

3.2.3 Cloning Design

1. Select the library blueprint or a discrete spacer sequence.
2. Insert your selection in the insulator sequence (see Subheading 3.1 for reference of the insulator sequence).
3. Assemble and clone the resulting promoter sequence upstream of RBS + ORF of choice.

3.3 Predicting TIF by Evaluation of Custom Spacers

As discussed in Subheading 3.1, TIF can be dependent on the local context, which is why the prediction task is only expected to work reliably if the insulator sequence (except spacer) is kept intact.

3.3.1 Using the Terminal

1. Open the terminal, activate the environment containing the packages described in Material.
2. Navigate to the directory containing the cloned GitHub repository.
3. Evaluate one or multiple spacer sequences by calling for example python ProD.py NNNCGGGNCCNGGGAAA AAACCCGGGTTTTAAAC. Alternatively, the sequence (s) can be provided as fasta file: python ProD.py ex_seqs.fa

4. The output is displayed in the terminal and additionally, a file my_predictions.csv is saved in the working directory. If another output path is desired, it can be specified in the previous step by adding -output_path PATH.

3.3.2 Using Jupyter

1. Open Jupyter in the environment containing the packages specified in Material and navigate to the directory containing the GitHub repository or open Binder via the ProD GitHub website.
2. Locate the 'Evaluate Custom Spacers' section at the bottom of the notebook.
3. Provide input data either as list input_data = ["AAAACCCCGGGGTTTTT", "TTTTCCCCGGGGTTTTT", "TTTTGGGGCCCCAAAAA"] or as file input_file = ["ex_seqs.fa"]
4. The output is displayed below the cell and additionally, a file my_predictions.csv is saved in the working directory. If another output path is desired, it can be specified in the previous step by adding output_path=PATH as argument for run_tool.

4 Notes

1. Binder is a service that can execute git repositories on the cloud for free. This can be achieved by visiting https://mybinder.org, or follow the link provided in the README. As such, no prerequisites are necessary. When using Binder, changes in the file are not saved.
2. The trained model is saved as model_RPOD.pt in the model folder and is used by the ProD tool to make predictions on novel input.
3. The 17 nucleotides sequence(s) are processed by a convolutional neural network of four 1×1, 16 1×4, and 32 1×2 convolutions and two fully connected layers of 128 and 64 nodes.
4. When evaluating the model, no specific nucleotide was found to occur with high frequency that could not be explained by the input blueprint. Therefore, we cannot confirm the findings of other studies that a single nucleotide has an effect on promoter TIF [12–14], but rather suggest the influence of more complex structural features.
5. When creating a library, the tool first creates all possible sequences from the blueprint input, second, it predicts the TIFs of each sequence and retains those matching the requested TIF. Spacers are then sampled to construct a

degenerate sequence for the library blueprint (unless the input allows for more than 500,000 possible sequences). If the input sequence for generating libraries allows for more than 500,000 possible sequences, the tool will randomly sample 100,000 possible sequences to evaluate. The processing time might be longer, and no library blueprint will be generated.

6. While the original insulator was successfully tested on plasmid and genome level, the in vivo validation of our tool was performed on plasmids.
7. The input DNA sequence may contain any letter of the mixed-base code (R Y M K S W H B V D N).

Acknowledgements

F.M. holds a PhD grant (61478) from FWO (Fonds Wetenschappelijk Onderzoek – Vlaanderen). This research was also supported by the BOF-IOP project "MLSB" (BOF16/ IOP/040) of the Bijzonder Onderzoeksfonds, the FWO research project "SynSys-Bio4COS" (G0B8118) of the Research Foundation – Flanders (FWO) and by the H2020 project "BioRoboost" of the European Union (820699). W.W. also received funding from the Flemish Government under the "Onderzoeksprogramma Artificielë Intelligentie (AI) Vlaanderen" Programme.

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